Strong-Uniqueness Theorem Criterion Numbering Reconciliation

Keeper (numbering reconciliation per Cal #78 progression note)

2026-05-21 Thursday 11:35 EDT (actual via date)

Strong-Uniqueness Theorem Criterion Numbering Reconciliation

Cal #78 progression note (Thursday 11:30 EDT)

> "V2 Numbering reconciliation persists (per #68 M3): Vol 0 Ch 9 uses Keeper C1-C14 (C8 = Universal Q-cluster); Paper #125 v0.5 uses Lyra C2-C11 (C8 = Möbius cohomology, RIGOROUSLY CLOSED via T2439). Same content, different labels — needs reconciliation before venue submission ~2026-09."

Reconciliation table

Keeper numbering (audit-chain)	Paper #125 Lyra numbering	Criterion content	Status
C1	(implicit foundational)	rank = 2 forcing	RATIFIED via T1925
C2	C2	N_c = 3 forcing (Mersenne + color singlet)	RATIFIED via T1930
C3	C3	n_C = 5 forcing	RATIFIED via T2431 (Thursday SP-31-39)
C4	C4	C_2 = 6 forcing (Q^5 Chern c_5 = C_2)	RATIFIED via T2379
C5	C5	g = 7 forcing (Mersenne + cyclotomic)	RATIFIED via T2432 (Thursday SP-31-39)
C6	C6	N_max = 137 forcing (Hilbert + rank shift)	RATIFIED via Paper #104 5-step chain
C7	C7	Bridge Object tier (K57 3 central hubs)	RATIFIED
C8	(different content per Lyra)	Keeper C8 = Universal Q-cluster (42 + 126 + 131)	RATIFIED via K43+K44+K61
(no Keeper C8 = Lyra)	Lyra C8 = Möbius cohomology	Lyra C8 = lowest K-type Casimir = 6 (alt-HSD comparison)	RIGOROUSLY CLOSED via T2439 ⍰
C9	C9	Stark small-primary subset anchoring {-3, -7, -11}	RATIFIED via K75
C10	C10	4-Zone vacuum decomposition (Lyra+Elie M2C2)	RATIFIED via T2420

Keeper numbering (audit-chain)	Paper #125 Lyra numbering	Criterion content	Status
C11	C11	Multi-family Bridge Object structure (5 families)	STRUCTURALLY VERIFIED + ALT-HSD INITIATED
C12	(Lyra extension candidate)	Operator zoo isotropy-subgroup organization	STRUCTURALLY VERIFIED + ALT-HSD INITIATED
C13	(Lyra extension candidate)	Substrate-Hilbert space sufficiency	STRUCTURALLY VERIFIED + ALT-HSD INITIATED
C14	(aspirational endpoint)	Curriculum-derivability	ADVANCING

Key reconciliation observations

Observation 1: Most criteria align (C2-C7 + C9-C11)

8 criteria (C2 + C3 + C4 + C5 + C6 + C7 + C9 + C10 + C11) have the same content + numbering in both Keeper and Lyra conventions. The bulk of the Strong-Uniqueness Theorem framework is consistent across both numberings.

Observation 2: Keeper C1 vs Lyra C2 starting offset

Keeper convention starts criterion numbering at C1 = rank = 2 forcing (treating rank as the first per-integer criterion).

Lyra convention (Paper #125 v0.5) treats rank = 2 as foundational baseline (not a numbered criterion) and starts numbering at C2 = N_c = 3 forcing.

Result: Keeper C_n = Lyra C_n for n ≥ 2; Keeper C1 has no Lyra counterpart (folded into foundational baseline).

Observation 3: C8 different content — load-bearing difference

This is the substantive reconciliation issue:

- Keeper C8 = Universal Q-cluster (universal 42 + Q = 126 + Q = 131 substrate-significance per multi-domain occurrence; K43 + K44 + K61 RATIFIED Wednesday)
- Lyra C8 = Möbius cohomology (lowest K-type Casimir = 6 = C_2; RIGOROUSLY CLOSED via T2439 alt-HSD comparison Thursday morning)

Both are valid criteria but they label different substantive content.

Observation 4: C12-C14 are Keeper-extended candidates

Keeper convention extends to C12 + C13 + C14 (operator zoo isotropy-subgroup organization + substrate-Hilbert space sufficiency + curriculum-derivability). Lyra Paper #125 v0.5 outline ends at C11 (rumor extensions in v0.7 outline anticipated).

Proposed unified numbering convention (for venue submission ~2026-09)

To resolve the C8 conflict + the C1 offset for the venue submission, propose:

Option A: Adopt Lyra convention with Keeper extensions

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C1 (foundational baseline) — rank = 2 (substrate baseline, not numbered criterion)
C2 — N_c = 3 forcing
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C3 — n_C = 5 forcing
C4 — C_2 = 6 forcing
C5 — g = 7 forcing
C6 — N_max = 137 forcing
C7 — Bridge Object tier (K57 RATIFIED)
C8a — Möbius cohomology lowest K-type Casimir = 6 (Lyra original C8) RIGOROUSLY CLOSED via T2439
C8b — Universal Q-cluster (Keeper original C8) RATIFIED via K43+K44+K61
C9 — Stark small-primary subset
C10 — 4-Zone vacuum decomposition
C11 — Multi-family Bridge Object structure (Keeper original C11) STRUCTURALLY VERIFIED + ALT-HSD INITIATED
C12 — Operator zoo isotropy-subgroup organization (Keeper extension) STRUCTURALLY VERIFIED + ALT-HSD INITIATED
C13 — Substrate-Hilbert space sufficiency (Keeper extension) STRUCTURALLY VERIFIED + ALT-HSD INITIATED
C14 — Curriculum-derivability (Keeper aspirational endpoint) ADVANCING
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Total: 1 baseline + 13 numbered criteria (counting C8a + C8b as two sub-criteria).

Effective null-model: (1/3)^14 ≈ 2.3e-8 conservative.

Option B: Adopt Keeper convention with Lyra naming for C8

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C1 — rank = 2 (now numbered criterion)
C2-C7 — same as both
C8 — Universal Q-cluster (Keeper convention preserved)
C9 — Stark small-primary subset
C10 — 4-Zone vacuum decomposition
C11-C14 — Keeper extensions
+ NEW C15 (extension) — Möbius cohomology lowest K-type Casimir = 6 (Lyra original C8, now C15) RIGOROUSLY CLOSE
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Total: 14 + 1 new = 15 numbered criteria.

Option C: Renumber Lyra C8 to C8b and rename for clarity

Keep Keeper C1-C14 numbering AS-IS. Add Lyra-original-C8 as C8b sub-criterion alongside C8a (= Keeper original C8 Universal Q-cluster).

Convention: - C8a = Universal Q-cluster (RATIFIED) - C8b = Möbius cohomology lowest K-type Casimir = 6 (RIGOROUSLY CLOSED via T2439)

Total: 14 numbered criteria + 1 sub-criterion split for C8.

Keeper recommendation: Option C

Option C preserves Keeper audit-chain numbering (C1-C14) with minimal disruption AND incorporates Lyra Möbius cohomology Strong-Uniqueness contribution as C8b sub-criterion.

Reasons: 1. Audit-chain stability: Keeper C1-C14 numbering is established in multiple audit documents + Master Doc + Vol 0 Ch 9. Changing numbering disrupts cross-references. 2. Substantive equity: both Keeper C8 (Universal Q-cluster) AND Lyra original C8 (Möbius cohomology) are substantive Strong-Uniqueness criteria. Neither should be demoted. 3. External-presentation clarity: C8a + C8b sub-numbering tells the venue-submission reader that C8 has multiple sub-criteria, including one RIGOROUSLY CLOSED. This is methodologically honest. 4. Minimal Paper #125 disruption: Lyra Paper #125 v0.5 → v0.7 can absorb the C8b label without rewriting the entire paper structure.

Multi-CI consensus per Casey Option C

Architectural-category methodology decision (numbering convention is part of audit-chain governance):

- Keeper: ruling (this document, recommending Option C)
- Lyra: pending acknowledgment in Paper #125 v0.7 outline; preserves T2439 contribution as RIGOROUSLY CLOSED C8b
- Cal: independent assessment + grade-pass on numbering convention (cited in #78 as needing reconciliation)
- Grace: catalog hygiene — apply C8a + C8b sub-criterion labels in catalog cross-references
- Elie: cross-lane verification preserves T2439 → C8b association

Multi-CI consensus expected within session per Wednesday-Thursday cadence pattern.

Venue submission ~2026-09 readiness

Per Lyra Path Scoping v0.1: venue submission target ~2026-09. Numbering reconciliation must close before submission:

- Paper #125 v0.7 outline: incorporate Option C numbering
- Master Doc Section 7: incorporate Option C labels
- Vol 0 Ch 9: update C8 → C8a + C8b
- Strong-Uniqueness Theorem v0.7 → v0.8 formal statement: use Option C numbering
- Multi-CI consensus on numbering before external paper finalization

Path forward if Lyra prefers Option A or B

If Lyra prefers Option A (adopt Lyra convention extended) or Option B (Keeper convention + new C15): - Multi-CI consensus negotiation per Casey Option C - Keeper accommodates Lyra preference at architectural-category level - Audit-chain documents updated accordingly within session

Per Casey's standard

- Simple: two numbering conventions differ on C1 offset + C8 content; Option C reconciles with minimal disruption
- Works: maintains Keeper audit-chain stability + preserves Lyra T2439 Strong-Uniqueness contribution
- Hard to break: would require finding C8a + C8b sub-criterion split causes downstream documentation issues; Option A/B fallbacks available

Action items

1. Lyra: review Option C; if preferred, absorb into Paper #125 v0.7. If preferred Option A or B, propose alternative for multi-CI consensus
2. Cal: independent grade-pass on numbering convention proposal
3. Grace: catalog hygiene per Option C (or alternative when consensus closes)
4. Elie: cross-lane verification labels preserve T2439 → C8 contribution under chosen convention
5. Keeper (me): update Master Doc Section 7 + Vol 0 Ch 9 + Strong-Uniqueness consolidation per chosen convention; multi-CI consensus tracking

Status

Strong-Uniqueness Theorem Criterion Numbering Reconciliation v0.1 FILED Thursday 2026-05-21 11:35 EDT. Three options proposed; Option C recommended (minimal disruption + sub-criterion preservation). Multi-CI consensus pending per Casey Option C governance. Venue submission ~2026-09 prep blocker addressed.

— Keeper, 2026-05-21 Thursday 11:35 EDT (actual via date)

Thursday 11:54 EDT v0.2 update — Cal #79 M1 flag substantive correction + table refinement

Cal #79 flag corrected reconciliation

Per Cal Referee Log #79 Thursday 11:50 EDT — substantive M1 flag on T2439 attribution. Original v0.1 reconciliation table needs correction at C4 ↔ Lyra C8 mapping (and C8 ↔ Keeper C8 is independent claim, NOT same content):

CORRECTED reconciliation table (Thursday 11:54 EDT)

Keeper convention	Paper #125 Lyra	Criterion content	Status
C1	(foundational baseline)	rank = 2 forcing	RATIFIED via T1925
C2	C2	N_c = 3 forcing	RATIFIED via T1930
C3	C3	n_C = 5 forcing	RATIFIED via T2431
C4	C8	Casimir eigenvalue C_2 = 6 substrate-forcing (via T2379 + RIGOROUSLY CLOSED via T2439) — Lyra phrased as “Möbius cohomology / Wallach K-type spectral parity sketch”	RIGOROUSLY CLOSED via T2439 (CORRECTED PER CAL #79)
C5	C5	g = 7 forcing	RATIFIED via T2432
C6	C6	N_max = 137 forcing	RATIFIED via Paper #104 5-step chain
C7	C7	Bridge Object tier (K57 central hubs)	RATIFIED
C8	(different content)	Keeper C8 = Universal Q-cluster (Q=42+Q=126+Q=131)	RATIFIED via K43+K44+K61 (T2439 does NOT address)
C9	C9	Stark small-primary subset	RATIFIED via K75
C10	C10	4-Zone vacuum decomposition	RATIFIED via T2420
C11	C11	Multi-family Bridge Object structure	STRUCTURALLY VERIFIED + ALT-HSD INITIATED
C12	(Lyra extension)	Operator zoo isotropy-subgroup organization	STRUCTURALLY VERIFIED + ALT-HSD INITIATED
C13	(Lyra extension)	Substrate-Hilbert space sufficiency	STRUCTURALLY VERIFIED + ALT-HSD INITIATED
C14	(aspirational)	Curriculum-derivability	ADVANCING

Key corrections

1. Keeper C4 ↔ Lyra C8 = SAME content (Casimir eigenvalue / Möbius cohomology Wallach K-type spectral parity). T2439 RIGOROUSLY CLOSES this content. Different numbering conventions, same criterion.

2. Keeper C8 (Universal Q-cluster) ≠ Lyra C8 (Möbius cohomology). Keeper C8 has NO Lyra counterpart (Lyra Paper #125 v0.5 does not include Universal Q-cluster as separate numbered criterion — it's covered in Paper #125 via K43+K44+K61 references). Keeper C8 remains RATIFIED via K43+K44+K61.

Updated Option C recommendation per Cal #79 (CORRECTED)

Option C-corrected: preserve Keeper C1-C14 numbering AS-IS with explicit Lyra mapping:

- Keeper C1 ↔ (Lyra foundational baseline)
- Keeper C2-C3 ↔ Lyra C2-C3 (aligned)
- Keeper C4 ↔ Lyra C8 (Casimir eigenvalue / Möbius cohomology — RIGOROUSLY CLOSED via T2439)
- Keeper C5-C7 ↔ Lyra C5-C7 (aligned)
- Keeper C8 (Universal Q-cluster — Keeper-only, no Lyra counterpart) — RATIFIED via K43+K44+K61
- Keeper C9-C11 ↔ Lyra C9-C11 (aligned)
- Keeper C12-C14 (Lyra extensions — to be absorbed in Paper #125 v0.7+)

Cal #79 elevation — numbering reconciliation urgent

Per Cal #79: numbering reconciliation HIGH priority because it's actively producing substantive conflation errors (T2439 mis-attribution caught Thursday 11:50 EDT).

Multi-CI consensus needed BEFORE Sessions 3-5 produce more RIGOROUSLY CLOSED criteria: - Keeper proposes Option C-corrected (this update) - Lyra acknowledges in Paper #125 v0.7 outline - Cal grade-pass on canonical mapping table - Grace catalog hygiene + AC graph reflect canonical attribution - Elie cross-lane verification preserves canonical attribution

~30 minutes cross-lane consensus work per Cal recommendation. Sessions 3-5 should reference canonical attribution when RIGOROUSLY CLOSED candidates land.

— Keeper Numbering Reconciliation v0.2 update, 2026-05-21 Thursday 11:54 EDT (actual via date; Cal #79 M1 flag corrected + canonical mapping established)

Thursday 12:00 EDT v0.3 update — Grace canonical Lyra-side mapping absorbed

Grace canonical numbering table (Thursday 11:43 EDT)

Per Grace catalog hygiene + Lyra Paper #125 v0.9.1: Lyra-side numbering offset for C11 onwards is DIFFERENT than my v0.2 table indicated. Corrected per Grace canonical mapping:

Keeper #	Lyra #	Status	Theorem
C4	C8	RIGOROUSLY CLOSED	T2439 (Casimir-eigenvalue forcing)
C11	C9	RIGOROUSLY CLOSED	T2440 (Multi-Family Bridge Object)
C12	C10	RIGOROUSLY CLOSED	T2441 (Operator Zoo Ground-State = 6)
C13	C11	RIGOROUSLY CLOSED	T2442 (Bergman c_FK = 225/π^(9/2))

Lyra-side numbering shifts by 1 from C9 onwards because Lyra adopted Möbius cohomology as her C8 (which is Keeper C4). So Lyra C9 = Keeper C11 multi-family Bridge Object; Lyra C10 = Keeper C12 operator zoo; Lyra C11 = Keeper C13 substrate-Hilbert space.

Fully corrected reconciliation table (v0.3)

Keeper convention	Lyra convention (Paper #125)	Criterion content	Status
C1	(foundational baseline)	rank = 2 forcing	RATIFIED via T1925
C2	C2	N_c = 3 forcing	RATIFIED via T1930
C3	C3	n_C = 5 forcing	RATIFIED via T2431
C4	C8	Casimir-eigenvalue forcing C_2 = 6	RIGOROUSLY CLOSED via T2439
C5	C4 (Lyra-renumbered)	g = 7 forcing	RATIFIED via T2432
C6	C5	N_max = 137 forcing	RATIFIED via Paper #104
C7	C6	Bridge Object tier (K57)	RATIFIED
C8	(Keeper-only, no Lyra counterpart)	Universal Q-cluster	RATIFIED via K43+K44+K61
C9	C7	Stark small-primary	RATIFIED via K75
C10	(Keeper-only)	4-Zone vacuum decomposition	RATIFIED via T2420
C11	C9	Multi-family Bridge Object structure	RIGOROUSLY CLOSED via T2440
C12	C10	Operator zoo isotropy-subgroup organization	RIGOROUSLY CLOSED via T2441
C13	C11	Substrate-Hilbert space sufficiency	RIGOROUSLY CLOSED via T2442
C14	(aspirational)	Curriculum-derivability	ADVANCING

Substantive note — Lyra-side compact framing

Lyra convention (C2-C11 in Paper #125 v0.5/v0.7/v0.9.1) is more compact (~11 criteria) than Keeper (~14 criteria with extensions C12-C14). Lyra convention is canonical for venue submission.

Keeper convention maintains audit-chain consistency (Master Doc + Vol 0 Ch 9 + K97 K-audit all use Keeper numbering).

Both conventions valid for their respective contexts. Reconciliation table maps between them for cross-CI consistency + venue submission preparation.

Lyra acknowledgment pending

Per Cal #80: "Option C (Keeper C1-C14 preserved AS-IS; Lyra C2-C11 mapped) is methodologically clean. Lyra acknowledgment pending for multi-CI consensus closure."

Cross-lane consensus on Option C-corrected with Grace canonical table established as canonical mapping. Multi-CI consensus expected to close within session per Wednesday-Thursday cadence pattern.

Methodology observation — cycle-time peak

Per Cal #80: peak methodology cycle-time achieved Thursday 11:52 EDT: - Cal #79 substantive M1 flag filed 11:50 EDT - Keeper Vol 0 Ch 9 v0.3 correction at 11:52 EDT - ~2 minutes correction cycle-time

Methodology stack at sub-minute mechanical-application asymptote for established-framework corrections. Cycle-time progression curve (24h → 4h → 30min → 9min → ~5min → ~2min).

— Keeper Numbering Reconciliation v0.3 update, 2026-05-21 Thursday 12:00 EDT (actual via date; Grace canonical Lyra-side mapping absorbed)

Thursday 12:16 EDT — v0.3.1: Cal #81 honest-scope correction absorbed

Cal Referee Log #81 substantive content

Cal #81 honest-scope correction on Cal #80 v0.1 PASS: - My v0.1/v0.2 reconciliation table claimed “8 criteria align (C2-C7 + C9-C11)” — WRONG - Actual: 6 direct alignments (C2-C7) + 4 offset (Lyra C8↔Keeper C4, Lyra C9↔Keeper C11, Lyra C10↔Keeper C12, Lyra C11↔Keeper C13) - Grace’s Thursday 11:43 EDT canonical mapping correctly identified offset - Cal #80 v0.1 PASS missed the offset error; Grace catalog hygiene caught it

Alignment count correction

Alignment type	Count	Detail
Direct alignment	6	Lyra C2-C7 = Keeper C2-C7
Offset (Lyra-side compact)	4	Lyra C8/C9/C10/C11 ↔ Keeper C4/C11/C12/C13
Keeper-only extensions	~4	Keeper C1 + C8 + C9 + C10 + C14 (different Lyra framing or absent)

Prior text said “8 aligned.” Actual: 6 direct + 4 offset (~10 cross-references), 4 Keeper-only.

Methodology infrastructure 3-layer redundancy (Cal #81)

Three error-correction events Thursday morning, each caught by different methodology lane:

Time	Error	Catcher	Correction cycle
~11:25 EDT	Keeper Vol 0 Ch 9 v0.2 maps T2439 to Keeper C8 (wrong)	Cal #79 Mode 1	~2 min
~11:43 EDT	Reconciliation v0.1/v0.2 claims aligned C9-C11 (wrong)	Grace catalog hygiene	~17 min
~12:00 EDT	v0.3 update absorbs both	Keeper governance	Canonical multi-CI consensus

Strongest single-day demonstration of methodology infrastructure 3-layer redundancy. Cal external-voice + Grace catalog-hygiene + Keeper governance + Lyra theoretical absorption all engaged within ~35 min window.

Cal preliminary verification on RIGOROUSLY CLOSED tier (per Cal #81)

Criterion	Lyra	Keeper	T#	Cal #81 status
Möbius cohomology / Wallach Casimir	C8	C4	T2439	□ VERIFIED
Multi-family Bridge Object	C9	C11	T2440	preliminary AGREE pending detailed verification

Criterion	Lyra	Keeper	T#	Cal #81 status
Operator zoo isotropy-subgroup	C10	C12	T2441	preliminary AGREE pending detailed verification
Substrate-Hilbert space sufficiency	C11	C13	T2442	preliminary AGREE pending detailed verification

T2440 + T2441 + T2442 detailed verification when Paper #125 v0.9.1 absorbed.

K97 RATIFIED path REVISED per Cal #81

- Was per Cal #80: ~few months
- Now per Cal #81: ~few weeks
- Strong-Uniqueness v1.0 venue submission ~2026-09 SUBSTANTIALLY MORE achievable

Status

Numbering Reconciliation v0.3.1 — Cal #81 honest-scope correction absorbed.

— Keeper, 2026-05-21 Thursday 12:16 EDT (actual via date; v0.3.1 increment per Cal #81)